

SECURE RHEL DEPARTMENTAL FILE MANAGEMENT AND AUTOMATED BACKUP SERVER

PROJECT REPORT

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Operating System: Red Hat Enterprise Linux 9.8
Platform: VMware
Project Disk: /dev/sdb – 20 GB
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ABSTRACT

This project implements a secure departmental file management and automated backup system on RHEL 9.8. A 20 GB disk was configured using LVM and divided into Finance, Operations, and Backup Logical Volumes. Users, groups and permissions were configured for controlled access. Backup and Rsync were performed using tar, and a Bash script was created to automate backups. Crontab schedules the backup daily at 8:00 PM. The backup and rysnc processes were successfully verified.

INTRODUCTION

This project focuses on creating a secure departmental file management and backup environment on RHEL 9.8. A dedicated disk was configured using LVM and divided into separate storage areas for Finance, Operations, and Backup. Users, groups, ownership, and permissions were configured to provide controlled access to departmental data.

The project also implements backup, rysnc, and automation using Linux tools. The tar utility was used to create and rysnc backups, while a Bash script was developed to automate the backup process. Crontab was configured to run the backup script daily at 8:00 PM, and the backup and restore processes were tested and verified successfully.

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1. Project Overview

This project demonstrates the implementation of a secure departmental file management and automated backup server using Red Hat Enterprise Linux 9.8.

A separate 20 GB disk was configured using LVM (Logical Volume Manager) and divided into three Logical Volumes:

- Finance
- Operations
- Backup

Linux users, groups, file permissions, departmental data management, backup and rync synchronization, Bash scripting and automated backup using Crontab.

The purpose of the project is to demonstrate practical Linux system administration in a controlled RHEL environment.

2. Project Objective

The main objectives of the project are:

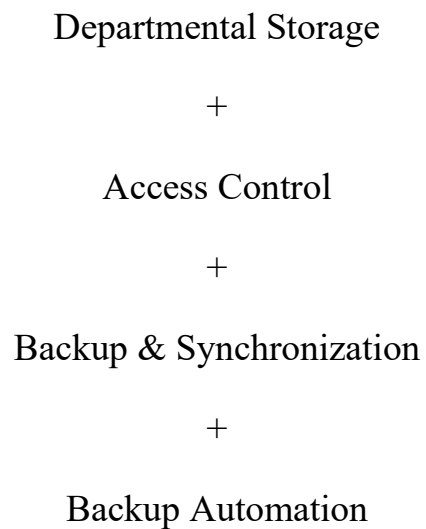
- To configure storage using LVM.
- To create separate storage areas for different departments.
- To format and mount the storage using XFS.
- To configure users and groups.
- To implement department-level file permissions.
- To create and manage departmental files.
- To implement backup and rsync.
- To develop a Bash backup script.
- To automate backups using crontab.
- To monitor and verify the Linux system.
- To troubleshoot common storage, permission and backup issues.

3. Problem Statement

In a departmental environment, different teams need separate storage areas for their files while maintaining controlled access.

There is also a requirement to protect important departmental data through regular backups.

The project addresses these requirements by creating using RHEL 9.8:



4. Project Scope

The scope of this project is to implement a secure departmental file management and automated backup system using RHEL 9.8. The project includes configuring a 20 GB disk using LVM and creating separate storage for Finance, Operations and Backup departments.

It also covers XFS filesystem creation, persistent mounting through `/etc/fstab`, LVM expansion, user and group management, file ownership and permissions and departmental data management.

For data protection, the project implements backup and rsync using tar, followed by the development of a Bash backup script. The backup process is automated using a crontab, scheduled to run daily at 8:00 PM.

Finally, the project includes testing, monitoring, verification and troubleshooting using Linux commands such as `lsblk`, `pvs`, `vgs`, `lvs`, `df`.

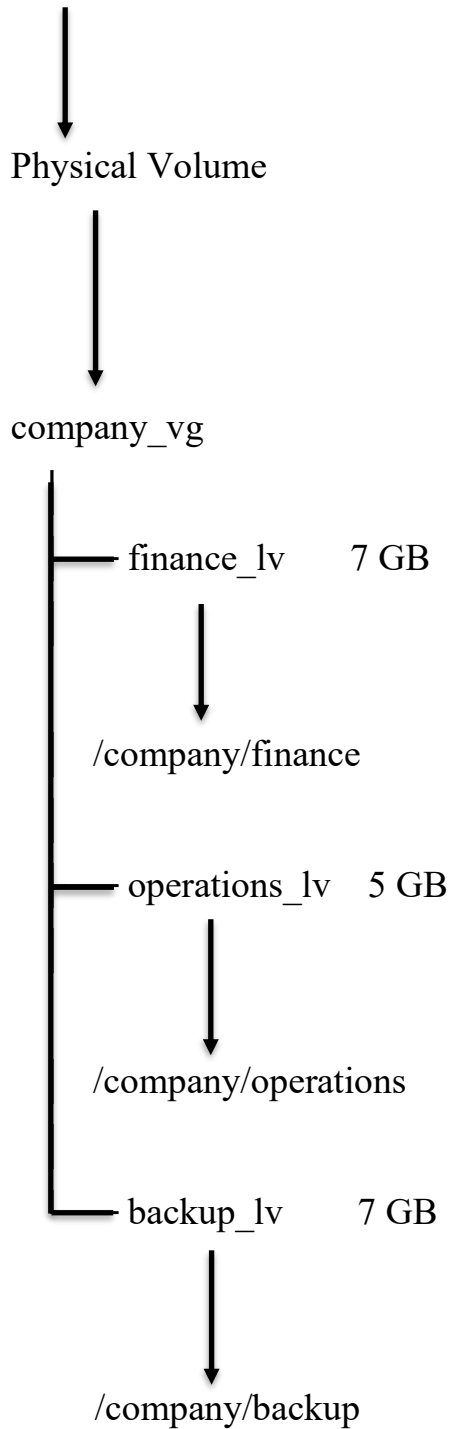
5. Technologies and Tools Used

SR.NO	TECHNOLOGY/TOOL	PURPOSE
1.	VMware	Virtual machine environment
2.	RHEL 9.8	Operating system
3.	LVM	Storage management
4.	XFS	Filesystem
5.	Bash	Backup Automation
6.	tar	Backup/archive creation
7.	Rsync	File Synchronization
8.	Crontab	Backup scheduling and monitoring
9.	crond	Cron job scheduling service
10.	vi	Configuration/file editing
11.	Linux users/groups	Access control

6. System Architecture

The project uses a separate 20 GB disk:

/dev/sdb



The final LVM configuration was:

PV: /dev/sdb

VG: company_vg

LVs:

finance_lv 7 GB

operations_lv 5 GB

backup_lv 7 GB

(Approximately 1 GB remained free in the Volume Group.)

7. Project Workflow

The project was implemented in the following sequence:

01 RHEL Setup RHEL 9.8 + VMware	02 Disk Identification /dev/sdb – 20 GB
03 LVM Configuration PV → VG → LV	04 Filesystem Configuration XFS
05 Mount Configuration Finance / Operations / Backup	06 User & Permission Management Users → Groups → Permissions
07 Departmental Data Finance + Operations	08 Backup & Synchronization tar → Rsync
09 Backup Automation Bash → Crontab	10 Monitoring & Testing

8. Linux Environment Setup

The project was implemented on:

OS: Red Hat Enterprise Linux 9.8

Platform: VMware

User: root

Project Disk: /dev/sdb

Disk Size: 20 GB

The operating system was verified using:

cat /etc/redhat-release

```
[root@localhost ~]# cat /etc/redhat-release
Red Hat Enterprise Linux release 9.8 (Plow)
```

Disk configuration was checked using:

lsblk

```
[root@localhost ~]# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0    0   20G  0 disk
├─sda1                              8:1    0   600M  0 part /boot/efi
├─sda2                              8:2    0    1G  0 part /boot
└─sda3                              8:3    0  18.4G  0 part
   ├─rhel_192-root                  253:0    0  16.4G  0 lvm  /
   └─rhel_192-swap                  253:1    0    2G  0 lvm  [SWAP]
sdb                                  8:16    0   20G  0 disk
├─company_vg-finance_lv            253:2    0    7G  0 lvm  /company/finance
├─company_vg-operations_lv         253:3    0    5G  0 lvm  /company/operations
└─company_vg-backup_lv             253:4    0    7G  0 lvm  /company/backup
sr0                                  11:0    1   232M  0 rom  /run/media/yogini/CDROM
sr1                                  11:1    1   14.5G  0 rom  /run/media/yogini/RHEL-9-8-0-BaseOS-x86_64
```

The existing OS disk was /dev/sda, while /dev/sdb was used as the dedicated project disk.

9. LVM Storage Configuration

9.1 Create Physical Volume

```
pvcreate /dev/sdb
```

(This initializes /dev/sdb as an LVM Physical Volume.)

Verification:

```
pvs
```

```
[root@localhost ~]# pvs
PV          VG          Fmt  Attr  PSize   PFree
/dev/sda3   rhel_192   lvm2 a--   18.41g     0
/dev/sdb    company_vg lvm2 a--   <20.00g 1020.00m
```

9.2 Create Volume Group

```
vgcreate company_vg /dev/sdb
```

(This creates the Volume Group)

Verification:

```
vgs
```

```
[root@localhost ~]# vgs
VG          #PV #LV #SN Attr   VSize   VFree
company_vg   1   3   0 wz--n- <20.00g 1020.00m
rhel_192     1   2   0 wz--n- 18.41g     0
```

9.3 Create Logical Volumes

Finance:

```
lvcreate -L 5G -n finance_lv company_vg
```

Operations:

```
lvcreate -L 5G -n operations_lv company_vg
```

Backup:

```
lvcreate -L 7G -n backup_lv company_vg
```

Later, Finance was expanded from 5 GB to 7 GB.

```
[root@localhost ~]# lvs
```

LV	VG	Attr	LSize	Pool
backup_lv	company_vg	-wi-ao----	7.00g	
finance_lv	company_vg	-wi-ao----	7.00g	
operations_lv	company_vg	-wi-ao----	5.00g	
root	rhel_192	-wi-ao----	16.41g	
swap	rhel_192	-wi-ao----	2.00g	

10. Filesystem and Mount Configuration

XFS was used for the Logical Volumes:

```
mkfs.xfs /dev/company_vg/finance_lv
```

```
mkfs.xfs /dev/company_vg/operations_lv
```

```
mkfs.xfs /dev/company_vg/backup_lv
```

Mount-point directories were created:

```
mkdir -p /company/finance
```

```
mkdir -p /company/operations
```

```
mkdir -p /company/backup
```

The filesystems were mounted:

```
mount /dev/company_vg/finance_lv /company/finance
```

```
mount /dev/company_vg/operations_lv /company/operations
```

```
mount /dev/company_vg/backup_lv /company/backup
```

Verification:

```
df -h | grep /company
```

```
[root@localhost ~]# df -h | grep /company
/dev/mapper//company_vg-backup_lv      7.0G   82M   6.9G    2% /company/backup
/dev/mapper//company_vg-operations_lv  5.0G   68M   4.9G    2% /company/operations
/dev/mapper//company_vg-finance_lv    7.0G   82M   6.9G    2% /company/finance
```

- **Persistent Mounting**

UUIDs were obtained using:

`blkid`

The UUIDs were added to:

`/etc/fstab`

Example:

`UUID=<FINANCE_UUID> /company/finance xfs defaults 0 0`

`UUID=<OPERATIONS_UUID> /company/operations xfs defaults 0 0`

`UUID=<BACKUP_UUID> /company/backup xfs defaults 0 0`

Then:

`systemctl daemon-reload`

`mount -a`

(This ensures the filesystems can be mounted automatically after a reboot.)

- **LVM Expansion**

Finance was expanded from 5 GB to 7 GB.

`lvextend -L +2G /dev/company_vg/finance_lv`

Because the filesystem was XFS:

`xfs_growfs /company/finance`

Verification:

`df -h /company/finance`

```
[root@localhost ~]# df -h /company/finance
Filesystem                                Size  Used Avail Use% Mounted on
/dev/mapper/company_vg-finance_lv        7.0G   82M   6.9G   2% /company/finance
```

11. User, Group and File Permission Management

Department-specific groups were created:

```
groupadd finance
```

```
groupadd operations
```

```
groupadd backup
```

Users were created:

```
useradd -m -G finance finance01
```

```
useradd -m -G operations operations01
```

```
useradd -m -G backup backup01
```

Passwords were assigned using:

```
passwd finance01
```

```
passwd operations01
```

```
passwd backup01
```

User membership was checked with:

```
id finance01
```

```
id operations01
```

```
id backup01
```

```
[root@localhost ~]# id finance01
uid=1001(finance01) gid=1004(finance01) groups=1004(finance01),1001(finance)
[root@localhost ~]# id operations01
uid=1003(operations01) gid=1006(operations01) groups=1006(operations01),1002(operations)
[root@localhost ~]# id backup01
uid=1005(backup01) gid=1008(backup01) groups=1008(backup01),1003(backup)
```

- **Ownership**

chown root:finance /company/finance

chown root:operations /company/operations

chown root:backup /company/backup

- **Permissions**

chmod 2770 /company/finance

chmod 2770 /company/operations

chmod 2770 /company/backup

(The **2** in 2770 enables the **setgid** bit.

This helps new files inherit the directory's group.)

```
[root@localhost ~]# ls -ld /company/*
drwxrws---. 2 root backup      230 Sep 24 08:25 /company/backup
drwxrws---. 2 root finance     57 Sep 21 10:46 /company/finance
drwxrws---. 2 root operations  40 Sep 13 14:54 /company/operations
```

12. Departmental File Management

Sample departmental files were created.

Finance:

/company/finance/annual_report_2026.txt

Operations:

/company/operations/operations_report_2026.txt

Backup:

/company/backup/backup_log.txt

The directory contents were checked using:

```
ls -lh /company/finance
```

```
ls -lh /company/operations
```

```
ls -lh /company/backup
```

```
[root@localhost ~]# ls -lh /company/*
/company/backup:
total 24K
-rw-rw----. 1 root backup 809 Sep 13 15:02 backup_log.txt
-rw-r--r--. 1 root backup 781 Sep 21 11:35 vaultflex_backup_2026-09-21.tar.gz
-rw-r--r--. 1 root backup 781 Sep 22 14:37 vaultflex_backup_2026-09-22.tar.gz
-rw-r--r--. 1 root backup 781 Sep 23 12:51 vaultflex_backup_2026-09-23.tar.gz
-rw-r--r--. 1 root backup 781 Sep 24 08:27 vaultflex_backup_2026-09-24.tar.gz
-rw-r--r--. 1 root backup 781 Sep 21 11:12 vaultflex_data_2026.tar.gz

/company/finance:
total 4.0K
-rw-rw----. 1 root      finance 866 Sep 13 14:50 annual_report_2026.txt
-rw-r--r--. 1 finance01 finance  0 Sep 21 10:46 test_file.txt

/company/operations:
total 4.0K
-rw-rw----. 1 root operations 764 Sep 13 14:54 operations_report_2026.txt
```

Access Testing

The Finance user was tested against the Finance directory.

```
su - finance01
```

Then:

```
ls -l /company/finance
```

Finance access was allowed.

- When the Finance user attempted to access Operations, the result was permission denied.
- **This demonstrated departmental access control.**

```
[root@localhost ~]# su - finance01
[finance01@localhost ~]$ ls -l /company/finance
total 4
-rw-rw----. 1 root      finance 866 Sep 13 14:50 annual_report_2026.txt
-rw-r--r--. 1 finance01 finance   0 Sep 21 10:46 test_file.txt
[finance01@localhost ~]$ ls -l /company/operations
ls: cannot open directory '/company/operations': Permission denied
```

13. Backup and Rsync Implementation

A manual backup was created using tar:

```
tar -czvf /company/backup/vaultflex_data_2026.tar.gz /company/finance  
/company/operations
```

tar options

- **c** = create
- **z** = gzip compression
- **v** = verbose
- **f** = specify archive file

The backup contents were verified:

```
tar -tzvf /company/backup/vaultflex_data_2026.tar.gz
```

```
[root@localhost ~]# tar -tzvf /company/backup/vaultflex_data_2026.tar.gz  
drwxrws--- root/finance      0 2026-09-21 10:46 company/finance/  
-rw-rw---- root/finance    866 2026-09-13 14:50 company/finance/annual_report_2026.txt  
-rw-r--r-- finance01/finance 0 2026-09-21 10:46 company/finance/test_file.txt  
drwxrws--- root/operations    0 2026-09-13 14:54 company/operations/  
-rw-rw---- root/operations  764 2026-09-13 14:54 company/operations/operations_report_2026.txt
```

Rsync File Synchronization

Rsync was used to synchronize the Finance departmental files with the backup location:

```
rsync -av /company/finance/ /company/backup/finance/
```

Rsync was also used to synchronize the Operations departmental files:

```
rsync -av /company/operations/ /company/backup/operations/
```

The synchronized files were verified using:

```
ls -lh /company/backup/finance/
```

```
ls -lh /company/backup/operations/
```

```
[root@localhost ~]# rsync -av /company/finance/ /company/backup/finance/
sending incremental file list

sent 122 bytes  received 12 bytes  268.00 bytes/sec
total size is 866  speedup is 6.46
[root@localhost ~]# rsync -av /company/operations/ /company/backup/operations/
sending incremental file list

sent 99 bytes  received 12 bytes  222.00 bytes/sec
total size is 764  speedup is 6.88
[root@localhost ~]# ls -lh /company/backup/finance/
total 4.0K
-rw-rw----. 1 root finance 866 Sep 13 14:50 annual_report_2026.txt
-rw-r--r--. 1 root finance   0 Sep 24 11:59 test_file.txt
[root@localhost ~]# ls -lh /company/backup/operations/
total 4.0K
-rw-r-----. 1 root operations 764 Sep 24 11:50 operations_report_2026.txt
[root@localhost ~]#
```

14. Backup Script Development

A Bash script was created:

`vi /usr/local/bin/vaultflex_backup.sh`

- The script performs the following tasks:
 1. Defines the backup location.
 2. Gets the current date.
 3. Creates a date-based backup filename.
 4. Archives Finance and Operations.
 5. Displays success or failure.

The script was made executable:

`chmod +x /usr/local/bin/vaultflex_backup.sh`

The backup script was tested manually using `/usr/local/bin/vaultflex_backup.sh`, which successfully created the backup file **`vaultflex_backup_2026-09-21.tar.gz`**.

```
[root@localhost ~]# /usr/local/bin/vaultflex_backup.sh
Starting VaultFlex backup...
Backup file: /company/backup/vaultflex_backup_2026-09-24.tar.gz
tar: Removing leading '/' from member names
tar: Removing leading '/' from hard link targets
Backup completed successfully.
```

15. Detailed Explanation of the Backup & Rsync Script

The script used in this project is:

```
#!/bin/bash

BACKUP_DIR="/company/backup"
DATE=$(date +%Y-%m-%d)
BACKUP_FILE="$BACKUP_DIR/vaultflex_backup_$DATE.tar.gz"

echo "Starting VaultFlex backup..."
echo "Backup file: $BACKUP_FILE"

# Create compressed Tar backup
tar -czf "$BACKUP_FILE" /company/finance /company/operations

if [ $? -eq 0 ]; then
    echo "Tar backup completed successfully."
else
    echo "Tar backup failed."
    exit 1
fi

# Synchronize Finance files using Rsync
echo "Starting Finance Rsync..."
rsync -av /company/finance/ /company/backup/finance/

if [ $? -eq 0 ]; then
    echo "Finance Rsync completed successfully."
```

```

else
    echo "Finance Rsync failed."
    exit 1
fi

# Synchronize Operations files using Rsync
echo "Starting Operations Rsync..."
rsync -av /company/operations/ /company/backup/operations/

if [ $? -eq 0 ]; then
    echo "Operations Rsync completed successfully."
else
    echo "Operations Rsync failed."
    exit 1
fi

echo "VaultFlex backup and synchronization completed successfully."

```

Explanation:

The vaultflex_backup.sh script automates the backup process by first creating a date-based compressed .tar.gz archive of the Finance and Operations directories. It then uses rsync -av to synchronize the Finance and Operations files into separate backup directories under /company/backup. After each operation, the script checks the exit status (\$?) to confirm whether the backup or synchronization was successful; if any operation fails, the script stops and reports the failure. Finally, the script displays a success message, and the Crontab executes this script automatically every day at **8:00 PM**.

16. Automated Scheduling using Crontab

The backup process was automated using **Crontab**. A cron job was configured to execute the VaultFlex Bash backup script automatically every day at **8:00 PM**.

The following entry was added to the root user's Crontab:

```
crontab -e
```

```
0 20 * * * /usr/local/bin/vaultflex_backup.sh
```

The configuration was verified using:

```
crontab -l
```

The Cron service was also verified using:

```
systemctl status crond
```

```
[root@localhost ~]# crontab -l
0 20 * * * /usr/local/bin/vaultflex_backup.sh
[root@localhost ~]# systemctl status crond
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: >
   Active: active (running) since Thu 2026-09-24 09:32:00 IST; 2h 8min ago
   Main PID: 1283 (crond)
     Tasks: 1 (limit: 10290)
    Memory: 1.2M (peak: 4.1M)
       CPU: 95ms
    CGroup: /system.slice/crond.service
           └─1283 /usr/sbin/crond -n

Sep 24 10:01:02 localhost.localdomain anacron[8239]: Anacron started on 2026-09-24
Sep 24 10:01:02 localhost.localdomain anacron[8239]: Will run job `cron.daily' at 10:01:02
Sep 24 10:01:02 localhost.localdomain anacron[8239]: Will run job `cron.weekly' at 10:01:02
Sep 24 10:01:02 localhost.localdomain anacron[8239]: Jobs will be executed sequentially
Sep 24 10:01:02 localhost.localdomain CROND[8225]: (root) CMDEND (run-parts /etc/cron.d)
Sep 24 11:26:40 localhost.localdomain anacron[8239]: Job `cron.daily' started
Sep 24 11:26:40 localhost.localdomain anacron[8239]: Job `cron.daily' terminated
Sep 24 11:26:40 localhost.localdomain anacron[8239]: Job `cron.weekly' started
Sep 24 11:26:40 localhost.localdomain anacron[8239]: Job `cron.weekly' terminated
Sep 24 11:26:40 localhost.localdomain anacron[8239]: Normal exit (2 jobs run)
```

The crond service was active and running, confirming that the scheduled backup could be executed automatically.

17. Logging, Monitoring and Verification

The backup process was monitored and verified using Linux commands.

The generated backup files were checked using:

```
ls -lh /company/backup
```

The contents of the backup archive were verified using:

```
tar -tzvf /company/backup/vaultflex_backup_2026-09-24.tar.gz
```

```
[root@localhost ~]# ls -lh /company/backup
total 24K
-rw-rw----. 1 root backup 809 Sep 13 15:02 backup_log.txt
-rw-r--r--. 1 root backup 781 Sep 21 11:35 vaultflex_backup_2026-09-21.tar.gz
-rw-r--r--. 1 root backup 781 Sep 22 14:37 vaultflex_backup_2026-09-22.tar.gz
-rw-r--r--. 1 root backup 781 Sep 23 12:51 vaultflex_backup_2026-09-23.tar.gz
-rw-r--r--. 1 root backup 768 Sep 24 12:00 vaultflex_backup_2026-09-24.tar.gz
-rw-r--r--. 1 root backup 781 Sep 21 11:12 vaultflex_data_2026.tar.gz
[root@localhost ~]# tar -tzvf /company/backup/vaultflex_backup_2026-09-24.tar.gz
drwxrws--- root/finance      0 2026-09-24 11:59 company/finance/
-rw-rw---- root/finance     866 2026-09-13 14:50 company/finance/annual_report_
2026.txt
-rw-r--r-- root/finance      0 2026-09-24 11:59 company/finance/test_file.txt
drwxrws--- root/operations   0 2026-09-24 11:50 company/operations/
-rw-r----- root/operations 764 2026-09-24 11:50 company/operations/operations_
report_2026.txt
```

This proves the backup archive contains the Finance and Operations data.

```
ls -lh /company/backup/finance/
```

```
ls -lh /company/backup/operations/
```

```
[root@localhost ~]# ls -lh /company/backup/finance/
ls -lh /company/backup/operations/
total 4.0K
-rw-rw----. 1 root finance 866 Sep 13 14:50 annual_report_2026.txt
-rw-r--r--. 1 root finance  0 Sep 24 11:59 test_file.txt
total 4.0K
-rw-r-----. 1 root operations 764 Sep 24 11:50 operations_report_2026.txt
```

18. Project Testing and Test Cases

Test	Expected Result	Result
RHEL version check	RHEL 9.8 displayed	Passed
LVM configuration	PV/VG/LVs available	Passed
Filesystem mount	All /company mounts available	Passed
Finance expansion	Finance reaches 7 GB	Passed
Department access	Authorized user accesses own directory	Passed
Unauthorized access	Permission denied	Passed
Backup creation	.tar.gz generated	Passed
Backup script	Script executes successfully	Passed
Crontab configuration	Daily 8 PM schedule displayed	Passed
Cron daemon status	crond active and running	Passed
Backup script	Script executes successfully	Passed
Backup file verification	.tar.gz file generated	Passed
Rsync Finance	Finance files synchronized	Passed
Rsync Operations	Operations files synchronized	Passed
Rsync Verification	Unchanged files not retransferred	Passed

19. Expected Output

The project successfully provides a secure departmental file management and automated backup environment on RHEL 9.8. Separate Finance, Operations, and Backup Logical Volumes are configured using LVM, with appropriate users, groups, ownership, permissions, and Setgid settings.

The backup process successfully creates compressed .tar.gz archives of the departmental data. Rsync synchronizes the Finance and Operations files with the backup directories. The backup and synchronization processes are automated through the Bash script and Crontab, scheduled to run daily at 8:00 PM. The generated backup archives, archive contents, and synchronized files were successfully verified.

```
[root@localhost ~]# tree /company
/company
├── backup
│   ├── backup_log.txt
│   ├── finance
│   │   ├── annual_report_2026.txt
│   │   └── test_file.txt
│   ├── operations
│   │   └── operations_report_2026.txt
│   ├── vaultflex_backup_2026-09-21.tar.gz
│   ├── vaultflex_backup_2026-09-22.tar.gz
│   ├── vaultflex_backup_2026-09-23.tar.gz
│   ├── vaultflex_backup_2026-09-24.tar.gz
│   └── vaultflex_data_2026.tar.gz
├── finance
│   ├── annual_report_2026.txt
│   └── test_file.txt
└── operations
    └── operations_report_2026.txt

5 directories, 12 files
```

20. Troubleshooting Guide

During the project implementation, Linux commands were used to identify and resolve issues related to storage, permissions, backup and automation. The backup script was executed manually to verify its operation, while `crontab -l` and `systemctl status crond` were used to verify the scheduled backup and Cron service. The `ls -lh /company/backup/` and `tar -tzvf` commands were used to verify the generated backup archive.

Rsync synchronization was verified using `rsync -av` and by checking the Finance and Operations backup directories. Running Rsync a second time confirmed that unchanged files were not retransferred. User, group, ownership and permission issues were checked using `id` and `ls -ld /company/*`.

21. Security Considerations

The project includes basic Linux access-control mechanisms:

- Separate departmental users.
- Separate departmental groups.
- Group ownership of directories.
- Restricted directory permissions.
- Setgid permissions.
- Department-level access testing.
- Separate backup storage.
- Backup verification and Rsync synchronization.

The permission model prevents users from freely accessing other departmental directories.

22. Advantages

The project provides:

- Structured storage management using LVM.
- Separate departmental storage.
- Flexible storage expansion.
- Controlled user access.
- Centralized backup storage.
- Automated backup.
- Backup verification.
- Rsync-based file synchronization.
- Crontab-based automated backup scheduling.
- Linux monitoring and troubleshooting experience.

23. Limitations

The current project is a **single-server lab environment** running inside VMware.

It does not currently include:

- Remote backup storage.
- Multiple backup servers.
- Backup encryption.
- Off-site disaster recovery.
- Automatic backup retention/cleanup.
- Email/SMS backup failure notifications.

24. Future Enhancements

Possible future enhancements include:

- **Remote Backup**

Send backups to another server or storage location.

- **Backup Retention**

Automatically retain a defined number of backups and delete older archives.

- **Encryption**

Encrypt sensitive backup archives.

- **Monitoring**

Add monitoring and alerts for:

- Disk usage
- Backup failures
- Service failures

- **Disaster Recovery**

Create a secondary backup location for recovery in case the primary server fails.

- **Improved Backup Management**

Add checksum verification and backup integrity checks.

25. Skills Demonstrated

Through this project, I demonstrated practical knowledge of:

- RHEL 9.8 administration
- Linux command line
- Disk and storage management
- LVM
- Physical Volumes
- Volume Groups
- Logical Volumes
- XFS
- Filesystem mounting
- /etc/fstab
- User management
- Troubleshooting
- Group management
- File ownership
- Linux permissions
- Setgid
- Bash scripting
- tar Backup
- Rsync File Synchronization
- Backup Verification
- Crontab Scheduling
- Cron (crond) Service Management
- Storage monitoring

26. Project Conclusion

The project successfully implemented a **secure departmental file management and automated backup environment on RHEL 9.8**. A dedicated 20 GB disk was configured using LVM with separate Finance, Operations and Backup Logical Volumes, along with XFS filesystems and departmental access permissions.

A backup process using tar was implemented, followed by a Bash backup script and Rsync-based file synchronization. The backup and synchronization process was automated using Crontab and scheduled to run daily at 8 PM. The project was tested for storage, permissions, backup, Rsync synchronization and automated scheduling.”